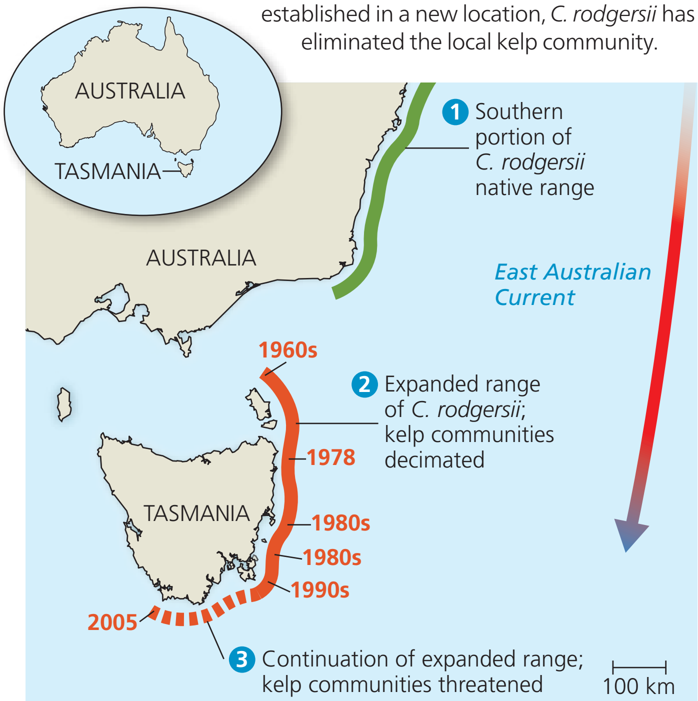
▼ Figure 52.21 A sea urchin's expanding range. Since the 1950s, water temperatures along the coast of Tasmania have increased, allowing the sea urchin *C. rodgersii* to expand its range to the south. The bold orange type indicates the years when *C. rodgersii* was first observed colonizing those locations. Once its population has been established in a new location, *C. rodgersii* has eliminated the local kelp community.



enable certain organisms, such as thermophilic prokaryotes, to live outside the temperature range habitable by other life.

As mentioned earlier, climate change has already caused hundreds of species to alter their geographic ranges. A shift in the range of one species can also have profound effects on the distribution of other species. Consider how rising sea temperatures have affected the geographic range of the sea urchin *C. rodgersii*. Since 1950, water temperatures along the coast of Tasmania, an island south of mainland Australia, have increased from 11.5°C to 12.5°C. This has enabled *C. rodgersii*—whose larvae fail to develop properly if temperatures drop below 12°C—to expand its range to the south **(Figure 52.21)**. The urchin is a voracious consumer of kelp and other algae. As a result, algal communities that once harbored a rich diversity of other species have been completely destroyed in regions where the urchin has become well established (denoted by a solid orange line in Figure 52.21).

Water and Oxygen

The dramatic variation in water availability among habitats is another important factor in species distribution. Species living at the seashore or in tidal wetlands can desiccate (dry out) as the tide recedes. Terrestrial organisms face a nearly constant threat of desiccation, and the distribution of terrestrial species reflects their ability to obtain and conserve water. Many amphibians, such as the tiny frog in Figure 52.1, are particularly vulnerable to drying because they use their moist, delicate skin for gas exchange. Desert organisms exhibit a variety of adaptations for acquiring and conserving water in dry environments, as described in Concept 44.4.

Water affects oxygen availability in aquatic environments and in flooded soils, where the slow diffusion of oxygen in water can limit cellular respiration and other physiological processes. Oxygen concentrations can be particularly low in deep ocean and deep lake waters, as well as in sediments where organic matter is abundant. Flooded wetland soils may also have low oxygen content. Mangroves and other trees have specialized roots that project above the water and help the root system obtain oxygen (see Figure 35.4). Unlike many flooded wetlands, the surface waters of streams and rivers tend to be well oxygenated because of rapid exchange of gases with the atmosphere.

Salinity

The salt concentration of water in the environment affects the water balance of organisms through osmosis. Most aquatic organisms are restricted to either freshwater or saltwater habitats by their limited ability to osmoregulate (see Concept 44.1). Most terrestrial organisms can excrete excess salts from specialized glands or in feces or urine. However, the salt concentrations in some habitats (such as salt flats) are so high that few species of plants or animals can survive there **(Figure 52.22)**. In the **Scientific Skills Exercise**, you can interpret data from an experiment investigating the influence of salinity on plant distributions.

Salmon that migrate between freshwater streams and the ocean use both behavioral and physiological mechanisms to osmoregulate. They balance their salt content by adjusting the amount of water they drink and by switching their gills from taking up salt in fresh water to excreting salt in the ocean.

Sunlight

Sunlight provides the energy that drives most ecosystems, and too little sunlight can limit the distribution of photosynthetic species. In forests, shading by leaves makes competition for light especially intense, particularly for seedlings growing on the forest floor. In aquatic environments, every meter of water depth absorbs about 45% of the red light and about 2% of the blue light passing through it. Thus, most photosynthesis occurs relatively near the water surface.

▼ Figure 52.22 Salar de Uyuni, in Bolivia, the largest salt flat in the world. Aside from the flamingos that breed there annually, very few animals or plants inhabit the vast expanse of crusty white salt.